

Year 7 • Chapter 5A • Algebra: answers (teacher copy)

5.01 The commutative and associative laws (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: $4 \times 12 \times 25 \rightarrow 1\ 200$
- Column 1, E2: $20 \times 6 \times 5 \rightarrow 600$
- Column 1, E3: $38 + 89 + 42 \rightarrow 169$
- Column 2, E1: $5 \times 15 \times 2 \rightarrow = 5 \times 2 \times 15; = 10 \times 15; = 150$
- Column 2, E2: $20 \times 4 \times 5 \rightarrow = 20 \times 5 \times 4; = 100 \times 4; = 400$
- Column 3, E1: $50 \times 9 \times 3 \times 2 \rightarrow = (50 \times 2) \times (9 \times 3); = 100 \times 27; = 2\ 700$
- Column 3, E2: $25 \times 11 \times 9 \times 4 \rightarrow = (25 \times 4) \times (11 \times 9); = 100 \times 99; = 9\ 900$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------|------------|------------|
| 1. 300 | 17. 179 | 33. 4 000 |
| 2. 400 | 18. 116 | 34. 18 000 |
| 3. 1 300 | 19. 143 | 35. 8 000 |
| 4. 280 | 20. 66 | 36. 1 200 |
| 5. 2 700 | 21. 111 | 37. 1 100 |
| 6. 1 000 | 22. 90 | 38. 1 500 |
| 7. 140 | 23. 97 | 39. 130 |
| 8. 1 900 | 24. 119 | 40. 3 000 |
| 9. 1 600 | 25. 153 | 41. 1 500 |
| 10. 1 000 | 26. 600 | 42. 4 500 |
| 11. 1 300 | 27. 160 | 43. 2 600 |
| 12. 500 | 28. 80 | 44. 960 |
| 13. 134 | 29. 50 | 45. 800 |
| 14. 131 | 30. 15 000 | 46. 6 000 |
| 15. 127 | 31. 700 | 47. 350 |
| 16. 154 | 32. 1 400 | 48. 4 500 |

5.02 The distributive law (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: $6 \times 26 \rightarrow 156$
- Column 1, E2: $5 \times 14 \rightarrow 70$
- Column 2, E1: $21 \times 40 \rightarrow = 20 \times 40 + 1 \times 40; = 800 + 40; = 840$
- Column 2, E2: $99 \times 27 \rightarrow = 100 \times 27 - 1 \times 27; = 2\ 700 - 27; = 2\ 673$
- Column 3, E1: $7(y - 5) \rightarrow = 7 \times y - 7 \times 5; = 7y - 35$
- Column 3, E2: $5(t + 9) \rightarrow = 5 \times t + 5 \times 9; = 5t + 45$

Questions (in order: Easy, Medium, Challenging)

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|---------|---------|---------------|
| 1. 174 | 17. 80 | 33. 294 |
| 2. 240 | 18. 423 | 34. 207 |
| 3. 164 | 19. 176 | 35. 798 |
| 4. 222 | 20. 217 | 36. 451 |
| 5. 96 | 21. 210 | 37. $6t - 30$ |
| 6. 144 | 22. 259 | 38. $6y - 30$ |
| 7. 235 | 23. 266 | 39. $3n + 3$ |
| 8. 144 | 24. 90 | 40. $9n + 9$ |
| 9. 84 | 25. 196 | 41. $5m - 5$ |
| 10. 126 | 26. 114 | 42. $9m + 9$ |
| 11. 231 | 27. 72 | 43. $3y + 6$ |
| 12. 215 | 28. 225 | 44. $6m + 18$ |
| 13. 256 | 29. 216 | 45. $3c - 3$ |
| 14. 175 | 30. 522 | 46. $7y - 84$ |
| 15. 126 | 31. 253 | |
| 16. 81 | 32. 841 | |

5.03 Variables and algebraic notation (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: $p \times p \rightarrow p^2$
- Column 1, E2: $b \times b \rightarrow b^2$
- Column 2, E1: $2bp \rightarrow = 2 \times b \times p; = 2pb$ (also correct)
- Column 2, E2: $7ap \rightarrow = 7 \times a \times p; = 7pa$ (also correct)
- Column 3, E1: $7 \times t \times y \times t \times 3 \rightarrow = 7 \times 3 \times t \times t \times y; = 21t^2y$
- Column 3, E2: $4 \times a \times c \times a \times 4 \rightarrow = 4 \times 4 \times a \times a \times c; = 16a^2c$

Questions (in order: Easy, Medium, Challenging)

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|-----------|---------------------------|---------------------------|
| 1. $3xy$ | 18. $10xy$ | 35. $7 \times k \times t$ |
| 2. $3cn$ | 19. n^2 | 36. $6 \times c \times k$ |
| 3. $8bx$ | 20. $3k$ | 37. $9 \times b \times k$ |
| 4. $5bp$ | 21. $2m$ | 38. $3 \times a \times t$ |
| 5. $12y$ | 22. $10p$ | 39. $2 \times m \times x$ |
| 6. $11n$ | 23. $6k$ | 40. $45b^2n$ |
| 7. $11ay$ | 24. $7k$ | 41. $16k^2n$ |
| 8. $11x$ | 25. $7y$ | 42. $20b^2y$ |
| 9. c^2 | 26. $6mx$ | 43. $36m^2x$ |
| 10. $12c$ | 27. $5n$ | 44. $15t^2y$ |
| 11. a^2 | 28. $8py$ | 45. $32m^2p$ |
| 12. t^2 | 29. $6m$ | 46. $15n^2p$ |
| 13. $6tx$ | 30. $8 \times b \times c$ | 47. $24c^2x$ |
| 14. $3x$ | 31. $3 \times k \times p$ | 48. $20m^2p$ |
| 15. $12t$ | 32. $2 \times c \times k$ | 49. $4t^2x$ |
| 16. $6n$ | 33. $6 \times b \times y$ | |
| 17. m^2 | 34. $8 \times b \times p$ | |

5.04 Expressions from words (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: 4 more than $a \rightarrow a + 4$
- Column 1, E2: 7 less than $a \rightarrow a - 7$
- Column 2, E1: 3 less than twice $k \rightarrow$ first: $2k; = 2k - 3$
- Column 2, E2: 11 more than 3 times $c \rightarrow$ first: $3c; = 3c + 11$
- Column 3, E1: the cost of m pens at \$15 each, plus \$18 for a folder $\rightarrow m$ = number of pens; cost = $15m + 18$ dollars
- Column 3, E2: the number of legs on x spiders and 13 birds $\rightarrow x$ = number of spiders; legs = $8x + 2 \times 13 = 8x + 26$
- Column 3, E3: \$100 shared equally by n people $\rightarrow n$ = number of people; each gets $\frac{100}{n}$ dollars

Questions (in order: Easy, Medium, Challenging)

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|-------------------|--------------------|----------------------------------|
| 1. $11p$ | 17. $9t$ | 33. $py - 7$ |
| 2. $\frac{c}{10}$ | 18. $\frac{n}{5}$ | 34. $2a - 8$ |
| 3. $c + 10$ | 19. $a + 9$ | 35. $7c + 6$ |
| 4. $2x$ | 20. $2b$ | 36. $5k - 2$ |
| 5. b^2 | 21. k^2 | 37. $a^2 + 11$ |
| 6. $c - 5$ | 22. $b - 8$ | 38. $ap - 9$ |
| 7. $x + 5$ | 23. $n + 7$ | 39. $2a - 4$ |
| 8. $x - 11$ | 24. $c - 2$ | 40. $5c + 11$ |
| 9. $4n$ | 25. $6b$ | 41. $15y + 18$ dollars |
| 10. $\frac{k}{7}$ | 26. $\frac{m}{12}$ | 42. $8t + 28$ |
| 11. $k + 8$ | 27. $k + 5$ | 43. $\frac{70}{p}$ dollars each |
| 12. $2c$ | 28. $2k$ | 44. $8b + 11$ dollars |
| 13. c^2 | 29. a^2 | 45. $8n + 8$ |
| 14. $c - 12$ | 30. $m - 4$ | 46. $\frac{130}{x}$ dollars each |
| 15. $a + 6$ | 31. $6b - 11$ | |
| 16. $k - 2$ | 32. $x^2 + 9$ | |

47. $9t + 12$ dollars

5.05 Substitution (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: $2n, n = 8 \rightarrow 16$
- Column 1, E2: $2a, a = 9 \rightarrow 18$
- Column 2, E1: $ab, a = 9, b = 5 \rightarrow 9 \times 5 = 45$
- Column 2, E2: $2a + b, a = 8, b = 3 \rightarrow 2 \times 8 + 3 = 19$
- Column 3, E1: Find A when $A = lw, l = 11, w = 4. \rightarrow A = lw; A = 11 \times 4; A = 44$
- Column 3, E2: Find P when $P = 2l + 2w, l = 3, w = 2. \rightarrow P = 2l + 2w; P = 2 \times 3 + 2 \times 2; P = 10$
- Column 3, E3: A plumber charges $C = 30h + 19$ dollars. Find C when $h = 3. \rightarrow C = 30h + 19; C = 30 \times 3 + 19; C = \109

Questions (in order: Easy, Medium, Challenging)

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|--------|--------|-----------------|
| 1. 8 | 16. 71 | 31. 17 |
| 2. 16 | 17. 17 | 32. 25 |
| 3. 32 | 18. 15 | 33. 22 |
| 4. 41 | 19. 64 | 34. 49 |
| 5. 28 | 20. 8 | 35. 40 |
| 6. 10 | 21. 55 | 36. 40 |
| 7. 18 | 22. 89 | 37. 81 |
| 8. 48 | 23. 49 | 38. 13 |
| 9. 36 | 24. 16 | 39. 6 |
| 10. 85 | 25. 8 | 40. 21 |
| 11. 24 | 26. 7 | 41. $A = 140$ |
| 12. 28 | 27. 25 | 42. $P = 28$ |
| 13. 20 | 28. 15 | 43. $C = \$139$ |
| 14. 41 | 29. 8 | 44. $A = 96$ |
| 15. 21 | 30. 13 | 45. $P = 40$ |

5.07 Adding and subtracting terms (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: $4y + 7y \rightarrow 11y$
- Column 1, E2: $11y + 3y \rightarrow 14y$
- Column 2, E1: $2k + 8p + 4k - 2p \rightarrow 2k + 4k + 8p - 2p = 6k + 6p$
- Column 2, E2: $8a + 5x + 3a - 4x \rightarrow 8a + 3a + 5x - 4x = 11a + x$
- Column 3, E1: $7tx + 8t^2 - 5tx + 5t^2 + 1 \rightarrow 7tx - 5tx + 8t^2 + 5t^2 + 1 = 2tx + 13t^2 + 1$
- Column 3, E2: $7nx + 9n^2 - 3nx + 5n^2 + 1 \rightarrow 7nx - 3nx + 9n^2 + 5n^2 + 1 = 4nx + 14n^2 + 1$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------|-----------------|-----------------------|
| 1. $9a$ | 18. $8x$ | 35. $6x + 5y$ |
| 2. $9p$ | 19. $2t$ | 36. $6b + 4y$ |
| 3. $6p$ | 20. $11p$ | 37. $17k + 2n$ |
| 4. $5a$ | 21. $11p$ | 38. $4b + t$ |
| 5. $8y$ | 22. $7t$ | 39. $12c + 9n$ |
| 6. $13n$ | 23. $5b$ | 40. $8ab + 11a^2 + 7$ |
| 7. $12b$ | 24. $12b$ | 41. $2ay + 6a^2 + 9$ |
| 8. $9m$ | 25. $17c$ | 42. $kn + 6k^2 + 6$ |
| 9. $10n$ | 26. $20c$ | 43. $3mx + 11m^2 + 7$ |
| 10. $6k$ | 27. $11y$ | 44. $-6cy + 9c^2 + 5$ |
| 11. $8k$ | 28. $15n$ | 45. $-mt + 12m^2 + 2$ |
| 12. $11y$ | 29. $5c$ | 46. $-4mx + 8m^2 + 4$ |
| 13. $8c$ | 30. $5a + 7b$ | 47. $5bx + 14b^2 + 9$ |
| 14. $19m$ | 31. $16m + 11t$ | 48. $-3bx + 7b^2 + 5$ |
| 15. $17x$ | 32. $12m + 6p$ | 49. $-3am + 6a^2 + 9$ |
| 16. x | 33. $18b - c$ | |
| 17. $12t$ | 34. $17n + 14t$ | |

5.08 Multiplying terms (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: $6k \times 8x \rightarrow 48kx$
- Column 1, E2: $5a \times 5m \rightarrow 25am$
- Column 2, E1: $6c \times 4c \rightarrow 6 \times 4 \times c \times c = 24c^2$
- Column 2, E2: $5by \times 5by \rightarrow 5 \times 5 \times by \times by = 25b^2y^2$
- Column 3, E1: $3m \times 3t \times 3m \rightarrow$ sign: +; numbers: $3 \times 3 \times 3 = 27$; letters: $m^2t = 27m^2t$
- Column 3, E2: $-2a \times (-5t) \times 3a \rightarrow$ sign: +; numbers: $2 \times 5 \times 3 = 30$; letters: $a^2t = 30a^2t$

Questions (in order: Easy, Medium, Challenging)

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|------------|--------------|---------------|
| 1. $36k$ | 22. $25cx$ | 42. $30k^2t$ |
| 2. $12px$ | 23. $36bn$ | 43. $10m^2t$ |
| 3. $20kt$ | 24. $27k$ | 44. $30k^2n$ |
| 4. $36c$ | 25. $4kt$ | 45. $8p^2$ |
| 5. $40a$ | 26. $21at$ | 46. $24mp$ |
| 6. $14bc$ | 27. $32n$ | 47. $12c^2y$ |
| 7. $56k$ | 28. $36n$ | 48. $24m^2$ |
| 8. $15b$ | 29. $48a$ | 49. $25p^2x$ |
| 9. $14bn$ | 30. $64kp$ | 50. $8bn$ |
| 10. $10k$ | 31. $4bp$ | 51. $24a^2$ |
| 11. $21am$ | 32. $6cp$ | 52. $18b^2m$ |
| 12. $64kn$ | 33. $20n^2$ | 53. $30k^2n$ |
| 13. $16ky$ | 34. $12ab^2$ | 54. $30m^2p$ |
| 14. $56ab$ | 35. $20a^2n$ | 55. $18m^2x$ |
| 15. $8m$ | 36. $36by^2$ | 56. $-20m^2p$ |
| 16. $28a$ | 37. $36cp$ | 57. $-75c^2m$ |
| 17. $35k$ | 38. $16m^2n$ | 58. $24b^2y$ |
| 18. $27n$ | 39. $30xy$ | 59. $-24b^2n$ |
| 19. $63ak$ | 40. $16a^2$ | |
| 20. $6p$ | 41. $20b^2$ | |
| 21. $40cm$ | | |

5.09 Dividing terms (booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: $18x \div 2 \rightarrow 9x$
- Column 1, E2: $24a \div 3 \rightarrow 8a$
- Column 2, E1: $12ny \div 4n \rightarrow \frac{12ny}{4n} = 3y$
- Column 2, E2: $18m \div 2m \rightarrow \frac{18m}{2m} = 9$
- Column 3, E1: $\frac{30m2y}{5my} \rightarrow$ numbers: $30 \div 5 = 6$; letters: $m^2y \div my = m; = 6m$
- Column 3, E2: $\frac{12a2x}{6ax} \rightarrow$ numbers: $12 \div 6 = 2$; letters: $a^2x \div ax = a; = 2a$

Questions (in order: Easy, Medium, Challenging)

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|----------|----------|----------|
| 1. $4b$ | 17. $8b$ | 33. 3 |
| 2. $7k$ | 18. $7m$ | 34. $6k$ |
| 3. $2b$ | 19. $5x$ | 35. $2a$ |
| 4. $8m$ | 20. $3t$ | 36. 9 |
| 5. $6n$ | 21. $7n$ | 37. 8 |
| 6. $5b$ | 22. $8x$ | 38. $4y$ |
| 7. $9a$ | 23. $6n$ | 39. $3y$ |
| 8. $8c$ | 24. $7p$ | 40. 9 |
| 9. $8t$ | 25. $8y$ | 41. 6 |
| 10. $4b$ | 26. $3t$ | 42. $9k$ |
| 11. $9k$ | 27. $6n$ | 43. $7b$ |
| 12. $8k$ | 28. $2p$ | 44. $4a$ |
| 13. $2k$ | 29. $5c$ | 45. 8 |
| 14. $3b$ | 30. $2a$ | 46. $9p$ |
| 15. $5n$ | 31. 4 | 47. $7b$ |
| 16. $8m$ | 32. $7p$ | 48. 2 |

49. 9

50. 9

51. 9*p*

52. 4*b*

53. 9*m*

54. 7*b*

55. 4*k*

56. 2*m*